

CASD empirical replication package — *The Minimum Wage in Firms' Organizations: Productivity Implications*

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Scope: CASD empirical component only. The JPE data exemption for this component was confirmed on **19 June 2026**; on **29 June 2026** the JPE Data Editor proposed use of **CASCAD certification** for this restricted-data component in place of the standard JPE replication procedure for that component. The structural-model solution and non-CASD simulations are outside this package.

This is the **single documentation file** for the package.

1. CASD users and computing environment

The restricted-data work was carried out by:

- **Claire Lelarge — Université Paris-Saclay;**
- **Grigorios Spanos — Université de Genève** at the time of the work.

The work was performed successively in CASD projects **COUTRAO** and **BROBOTS**. Both used **CONFIG3: 4 vCPU, 16 GB RAM, 600 GB disk**.

The certification-run hardware, operating system, SAS version, Stata version and user-written Stata package versions must be recorded after the successful CASCAD run in Section 5.

CASD/CASCAD procedure: <https://www.casd.eu/le-centre-dacces-securise-aux-donnees-casd/certification-de-resultats-cascad-casd/>

JPE Data Editor guidance: <https://jpedataeditor.github.io/>

2. Package contents

- code/manual/ — **author-tested execution path**.
- code/Programs/ — SAS/Stata scientific programs needed by the retained manual path.
- code/Tools/ — only the three helpers needed by the retained path: SAS environment check, Stata environment check, and CASD path resolver.
- data/public/ — public/third-party auxiliary inputs needed by the code; rights status is in data/public/PUBLIC_DATA_MANIFEST.csv.
- output/EXHIBIT_OUTPUT_MAP.csv — exhibit → generated file → program mapping.
- manuscript/paper.pdf — certification copy required for CASCAD/CASD review.
- LICENSE*.txt — package licensing terms.

3. Data Availability Statement and rights

Restricted CASD data

No confidential CASD microdata or confidential generated analysis datasets are distributed in this archive. Replicating results requires an authorized CASD environment containing the DOI-identified products in Section 4. Set:

- CASD_DATA_ROOT = the root of the authorized CASD project data area;
- REPLICATION_ROOT = the absolute path to this unzipped package.

The code resolves CASD subdirectory names at run time and does not hard-code a named user or project. Independent researchers must obtain their own CASD/data-producer authorization. Access time and cost depend on the applicable authorization process and current CASD tariffs. For the CASCAD certification procedure itself, the CASD page currently states a normal duration of **15–30 days** and lists the applicable certification/environment charges; users should verify current terms directly with CASD/CASCAD.

JPE exemption

The JPE data exemption applies to the CASD-relevant component only. The CASCAD route is the agreed/proposed verification mechanism for this restricted-data component.

Public auxiliary inputs

The 18 auxiliary inputs listed in `data/public/PUBLIC_DATA_MANIFEST.csv` are included here because they are needed by the retained replication path. They are third-party materials and are **not relicensed by the authors**. Every row is currently marked **“REUSE AUTHORIZATION IS NECESSARY”**.

Rights statements

The authors state that they had legitimate access to and permission to use the CASD data for the research.

Author-owned code is BSD-3-Clause; author-created documentation/metadata is CC BY 4.0. Confidential CASD data, third-party inputs and the manuscript are excluded from those grants; see `LICENSE.txt`.

4. CASD data citations and DOI versions

The following **42 CASD product/version identifiers** define the restricted inputs used by this package.

Source	Year	Producer	DOI
BTS-Postes / DADS Postes	1995	INSEE	10.34724/CASD.21.40.V1
BTS-Postes / DADS Postes	1996	INSEE	10.34724/CASD.21.41.V1
BTS-Postes / DADS Postes	1997	INSEE	10.34724/CASD.21.42.V1
BTS-Postes / DADS Postes	1998	INSEE	10.34724/CASD.21.43.V1
BTS-Postes / DADS Postes	1999	INSEE	10.34724/CASD.21.44.V1
BTS-Postes / DADS Postes	2000	INSEE	10.34724/CASD.21.45.V1
BTS-Postes / DADS Postes	2001	INSEE	10.34724/CASD.21.46.V1
BTS-Postes / DADS Postes	2002	INSEE	10.34724/CASD.21.47.V1
BTS-Postes / DADS Postes	2003	INSEE	10.34724/CASD.21.48.V1
BTS-Postes / DADS Postes	2004	INSEE	10.34724/CASD.21.49.V1
BTS-Postes / DADS Postes	2005	INSEE	10.34724/CASD.21.50.V1
BTS-Postes / DADS Postes	2006	INSEE	10.34724/CASD.21.51.V1
BTS-Postes / DADS Postes	2007	INSEE	10.34724/CASD.21.52.V1
BTS-Postes / DADS Postes	2008	INSEE	10.34724/CASD.21.53.V1
BTS-Entreprises / DADS Entreprises	1993	INSEE	10.34724/CASD.91.772.V1
FICUS / SUSE	1996	INSEE	10.34724/CASD.68.773.V1
FICUS / SUSE	1997	INSEE	10.34724/CASD.68.774.V1
FICUS / SUSE	1998	INSEE	10.34724/CASD.68.775.V1
FICUS / SUSE	1999	INSEE	10.34724/CASD.68.776.V1
FICUS / SUSE	2000	INSEE	10.34724/CASD.68.777.V1

Source	Year	Producer	DOI
FICUS / SUSE	2001	INSEE	10.34724/CASD.68.778.V1
FICUS / SUSE	2002	INSEE	10.34724/CASD.68.779.V1
FICUS / SUSE	2003	INSEE	10.34724/CASD.68.553.V1
FICUS / SUSE	2004	INSEE	10.34724/CASD.68.554.V1
FICUS / SUSE	2005	INSEE	10.34724/CASD.68.555.V1
FICUS / SUSE	2006	INSEE	10.34724/CASD.68.556.V1
FICUS / SUSE	2007	INSEE	10.34724/CASD.68.557.V1
FARE / ESANE	2010	DGFIP / INSEE	10.34724/CASD.42.124.V1
FARE / ESANE	2011	DGFIP / INSEE	10.34724/CASD.42.860.V1
FARE / ESANE	2012	DGFIP / INSEE	10.34724/CASD.42.1026.V1
FARE / ESANE	2013	DGFIP / INSEE	10.34724/CASD.42.1027.V1
FARE / ESANE	2014	DGFIP / INSEE	10.34724/CASD.42.1268.V1
FARE / ESANE	2015	DGFIP / INSEE	10.34724/CASD.42.2076.V2
FARE / ESANE	2016	DGFIP / INSEE	10.34724/CASD.42.2697.V1
Déclarations 2483	2002	Ministry of Labour (DARES)	10.34724/CASD.144.2243.V1
Déclarations 2483	2003	Ministry of Labour (DARES)	10.34724/CASD.144.2244.V1
Déclarations 2483	2004	Ministry of Labour (DARES)	10.34724/CASD.144.2245.V1
Déclarations 2483	2005	Ministry of Labour (DARES)	10.34724/CASD.144.1918.V1
Déclarations 2483	2006	Ministry of Labour (DARES)	10.34724/CASD.144.1919.V1
Déclarations 2483	2007	Ministry of Labour (DARES)	10.34724/CASD.144.1920.V1
Allègement des cotisations sociales	2000	Ministry of Labour (DARES)	10.34724/CASD.200.1917.V1
BIC-RN	2008	DGFIP / fiscal data	10.34724/CASD.259.2295.V1

CASD acknowledgement recommended by CASD for English-language publications: “Access to some confidential data, on which this work is based, was made possible within a secure environment offered by CASD – Centre d’accès sécurisé aux données, the French Centre for Secure Access to Data (Ref. 10.34724/CASD).”

5. Software, packages, hardware and expected runtime

Required software: **SAS** and **Stata**. Stata source files use version 17.0 syntax compatibility. Required user-written Stata commands are: `esttab`, `estpost`, `estout`, `tabout`, `reghdfe`, `ivreghdfe`, `ivreg2`, `ranktest`, `ftools`. The package does not install or update them automatically.

Run `code/manual/01_SAS_CHECK_ENVIRONMENT.sas` and `code/manual/02_STATA_CHECK_ENVIRONMENT.do` before scientific stages. They create environment records under `output/run_metadata/`.

Item	Value
Historical CASD hardware	CONFIG3 — 4 vCPU, 16 GB RAM, 600 GB disk
Historical -run OS	Windows Server 2019 Standard, version 1809, set-up date: 03/08/2023, OS build: 17763.9121
Historical t SAS version	SAS version 9.04.01M7P08052020, for WINDOWS (X64_SRV19), 4CPU.
Historical Stata version/edition/update	STATA 17, for Windows (PC 64-bit x86-64), MP 4 processors, born date: 18 Jul 2023.
<code>esttab</code> / <code>estpost</code> / <code>estout</code> versions	2.1.1 / 1.2.2 / 3.31
<code>tabout</code> version	2.0.8
<code>reghdfe</code> version	6.12.3
<code>ivreghdfe</code> version	1.1.3

Item	Value
ivreg2 version	4.1.11
ranktest version	2.0.04
ftools version	2.49.1

Expected runtime: manual wrappers write timing files to output/logs/timings/.

Stage	Task	Expected runtime
00	Stata setup	0 seconds
01	SAS environment check	0.031 seconds
02	Stata environment check	0 seconds
03	RTT/GMR assignments	3.277 seconds
04	DADS jobs	5,322.532 seconds
05	minimum-wage inputs	129.273 seconds
06	FICUS accounts	124.175 seconds
07	2483 training data	2.706 seconds
08	DADS 1993 calibration	7.915 seconds
09	industry nomenclatures	21.085 seconds
10	Appendix G / Table G1 FARE moment	151.208 seconds
11	Stata preprocessing	30 seconds
12A	organization base	1,052.828 seconds
12B	annual blocks 1996–2008	85.977 + 88.568 + 94.308 + 100.383 + 115.976 + 127.205 + 137.995 + 134.464 + 138.247 + 136.758 + 157.899 + 152.265 + 151.175 = 1,621.220 seconds
12C	organization finalize	842.953 seconds
13	final analysis data	497.000 seconds
14	empirical analysis	698.000 seconds
15	Table B1	64.000 seconds
TOTAL	complete CASD certification scope	10,568.203 seconds = 2 h 56 min 8.203 s

6. Author-tested execution procedure

The sequence of stages is implemented in the two files: "<ROOT>/code/Manual_mode_STATA_v1.do" and "<ROOT>/code/Manual_mode_SAS_v1.do"

6.1 Initialize Stata

```
do "<ROOT>/code/manual/00_PREPARE_DIRECTORIES.do" "<ROOT>" "<CASD_DATA_ROOT>"
```

This defines \$root, \$casd_data_root, creates runtime output directories and initializes timing files. If Stata is restarted, rerun this command.

6.2 Initialize SAS

At the start of each SAS session:

```
%let root=<ROOT>;
%let casd_data_root=<CASD_DATA_ROOT>;
%let keep_intermediate=1;
```

6.3 Run order

Stage	Software	File
00	Stata	code/manual/00_PREPARE_DIRECTORIES.do
01	SAS	code/manual/01_SAS_CHECK_ENVIRONMENT.sas
02	Stata	code/manual/02_STATA_CHECK_ENVIRONMENT.do
03	SAS	code/manual/03_SAS_BUILD_RTT_GMR_ASSIGNMENTS.sas
04	SAS	code/manual/04_SAS_EXTRACT_DADS_JOBS.sas
05	SAS	code/manual/05_SAS_BUILD_MINIMUM_WAGE_INPUTS.sas
06	SAS	code/manual/06_SAS_EXTRACT_FICUS_ACCOUNTS.sas
07	SAS	code/manual/07_SAS_EXTRACT_TRAINING_2483.sas
08	SAS	code/manual/08_SAS_EXTRACT_DADS_1993_CALIBRATION.sas
09	SAS	code/manual/09_SAS_EXTRACT_INDUSTRY_NOMENCLATURES.sas
10	SAS	code/manual/10_SAS_TABLE_G1_ZERO_LAYER_MOMENT.sas
11	Stata	code/manual/11_STATA_PREPROCESSING.do
12	SAS	<pre>%let root=<ROOT>; %let casd_data_root=<CASD_DATA_ROOT>; %let keep_intermediate=0; %include "&root\code\manual\low_resource\12A_SAS_BUILD_ORGANIZATION_BASE.sas"; Then run 12B_1996.sas through 12B_2008.sas, one annual block at a time, followed by: %include "&root\code\manual\low_resource\12C_SAS_BUILD_ORGANIZATION_FINALIZE.sas"; SAS may be closed between annual blocks to release WORK resources. After reopening, reset root, casd_data_root, and keep_intermediate.</pre>
13	Stata	code/manual/13_STATA_BUILD_FINAL_ANALYSIS_DATA.do
14	Stata	code/manual/14_STATA_EMPIRICAL_ANALYSIS.do
15	Stata	code/manual/15_STATA_TABLE_B1.do

For a SAS stage, run `%include "&root\code\manual\<file>.sas";`. For a Stata stage, run `do "$root/code/manual/<file>.do"`.

Purge SAS log files after each step.

6.4 Recovery after an error

If a manual stage ends in error, **close SAS or Stata completely and reopen it before rerunning the failed stage**. After restarting Stata, rerun Stage 00; after restarting SAS, redefine the three SAS macro variables above. Rerun the failed stage from its beginning and do not continue downstream until it completes cleanly. This also applies to 12A, any 12B_YYYY, and 12C.

7. Outputs and exhibit mapping

Generated material is written under `output/generated/`; logs/timings under `output/logs/`; environment records under `output/run_metadata/`. Confidential generated datasets under `output/generated/Output_data/` must remain inside CASD unless separately cleared for export.

The complete mapping from each certified exhibit to generated output and program is in `output/EXHIBIT_OUTPUT_MAP.csv`. The manuscript is the comparison reference; duplicated reference-output copies are intentionally not included in this lean version.

8. Variable glossary

The CASD/CASCAD guidance asks for a glossary when paper concepts and code variable names differ. The glossary is included directly below so that README.md remains the single documentation point.

Paper concept	Code variable	Definition / allowed values	Construction/source
Firm identifier	siren	SIREN firm identifier (9 characters). (identifier)	Core merge key across stages
Panel identifier	gsiren	Numeric group identifier generated from SIREN for Stata panel declaration. (integer identifier)	egen group(siren)
Calendar year	year	Calendar year of firm observation. (year)	Annual panel index
GMR regime	gmr	Garantie Mensuelle de Rémunération regime assigned from RTT agreement timing. Values 1-6; baseline regression sample excludes GMR 6. (categorical 1-6)	RTT/GMR assignment pipeline
RTT agreement year	annee_rtt	Year of the firm workweek-reduction agreement used to determine the GMR regime. (year)	01_build_rtt_gmr_assignments.sas
Agreement type	typeacc	Aubry/RTT agreement category. Category 9 is excluded from retained empirical samples. (categorical)	code/Programs/1_SAS_DADS/01_build_rtt_gmr_assignments.sas
Statutory minimum wage	min_wage	Applicable statutory minimum wage measure carried into the firm-year file. (euros, schedule-specific)	macro_build_firm_organization_data.sas
Hourly statutory minimum wage	min_wage_h	Hourly statutory minimum wage. (euros/hour)	macro_build_firm_organization_data.sas
Total labor cost at minimum wage	s_supbrut_mwh	Employer total hourly labor cost at the applicable minimum wage after payroll contributions/reliefs. (euros/hour)	macro_compute_labor_costs.sas; macro_build_firm_organization_data.sas
Log labor cost at minimum wage	lnsupmwh	Natural log of s_supbrut_mwh. (log euros/hour)	fn_apply_analysis_sample_filters.do
Three-year change in labor cost	lns_s3	Three-year difference S3.lnsupmwh; in 2006 this is the 2003-2006 change. (log difference)	fn_apply_analysis_sample_filters.do
2002 low-wage exposure	sharelowwagehrs02	Firm share of 2002 hours paid at or below the submitted 2006 minimum-wage thresholds; propagated to all firm-years. (share [0,1])	fn_apply_analysis_sample_filters.do; macro_build_firm_organization_data.sas
Hierarchy layers	layerlenient	Retained firm hierarchy-layer count after recoding. In data_lp02_analysis it takes 1-4 and includes the production layer; descriptive Table B code subtracts one to report management layers above production. (integer 1-4)	macro_build_firm_organization_data.sas; fn_assemble_annual_firm_panel.do
Layer indicators	layer1-layer4	Indicators for firms with exactly 1, 2, 3, or 4 retained total layers in data_lp02_analysis. (binary)	fn_apply_analysis_sample_filters.do
Change in layers: increase	incl	Indicator that S3.layerlenient > 0. (binary)	fn_apply_analysis_sample_filters.do
Change in layers: decrease	decl	Indicator that S3.layerlenient < 0. (binary)	fn_apply_analysis_sample_filters.do
Change in layers: unchanged	same1	Indicator that S3.layerlenient = 0. (binary)	fn_apply_analysis_sample_filters.do
Jobs / employment	totpostes	Number of retained DADS job records for the firm-year. (jobs)	macro_build_firm_organization_data.sas
Log jobs	Intotpostes	Natural log of total retained jobs across layers. (log jobs)	fn_apply_analysis_sample_filters.do
Layer jobs	totpostes1-totpostes4	Retained job counts by hierarchy layer after compacting empty occupation layers. (jobs)	fn_apply_analysis_sample_filters.do
Log layer jobs	Intotpostes1-Intotpostes4	Natural logs of layer-specific retained job counts. (log jobs)	fn_apply_analysis_sample_filters.do
Hours	tothrs	Total retained paid hours across hierarchy layers. (hours/year)	macro_build_firm_organization_data.sas
Full-time-equivalent employment	tothrs_fte	Total hours divided by 1607. (FTE)	01_build_descriptive_tables.do
Log hours	Intothrs	Natural log of total retained hours across layers. (log hours)	fn_apply_analysis_sample_filters.do
Layer hours	tothrs1-tothrs4	Retained hours by hierarchy layer after compacting empty layers. (hours/year)	fn_apply_analysis_sample_filters.do
Log layer hours	Intothrs1-Intothrs4	Natural logs of layer-specific hours. (log hours)	fn_apply_analysis_sample_filters.do
Average hourly labor cost	avsupbruthrs	Total employer labor cost divided by retained hours. (euros/hour)	fn_apply_analysis_sample_filters.do

Paper concept	Code variable	Definition / allowed values	Construction/source
Log average hourly labor cost	lnavsupbruthrs	Natural log of total employer labor cost divided by retained hours. (log euros/hour)	fn_apply_analysis_sample_filters.do
Layer hourly labor costs	avsupbruthrs1- avsupbruthrs4	Employer labor cost per hour by hierarchy layer. (euros/hour)	fn_apply_analysis_sample_filters.do
Log layer hourly labor costs	lnavsupbruthrs1- lnavsupbruthrs4	Natural logs of layer-specific employer labor cost per hour. (log euros/hour)	fn_apply_analysis_sample_filters.do
Training participation, production layer	D_h_fp_L0	Change/indicator derived from Déclarations 2483 training hours for the production layer (L0), using the submitted fallback-year construction. (indicator / change)	01_prepare_training_2483.do
Training participation, management layers	D_h_fp_L123	Change/indicator derived from Déclarations 2483 training hours for management layers L1-L3. (indicator / change)	01_prepare_training_2483.do
Training sample indicator	CONT_FP	Firm has nonmissing retained training measures in the 2006 training merge. (binary)	00_run_regression_analysis.do
Value added	vaht	Firm value added measure from FICUS/SUSE; positive nonmissing values required through 2007. (currency, source units)	fn_apply_analysis_sample_filters.do
Log value added	lnva	Natural log of vaht. (log currency)	fn_apply_analysis_sample_filters.do
Sales	catotal	Firm total sales/turnover from FICUS/SUSE. (currency, source units)	fn_apply_analysis_sample_filters.do
Log sales	lnsales	Natural log of catotal. (log currency)	fn_apply_analysis_sample_filters.do
Capital stock	immocor	Firm fixed assets/capital measure from FICUS/SUSE; positive nonmissing values required through 2007. (currency, source units)	fn_apply_analysis_sample_filters.do
Intermediate inputs	inputs	achat_mp + achat_mar + achat_serv. (currency, source units)	fn_apply_analysis_sample_filters.do; fn_estimate_tfp.do
Investment	invcorp	Firm investment measure carried from FICUS/SUSE. (currency, source units)	fn_apply_analysis_sample_filters.do
Value added per worker	lnvapw	Natural log of vaht/totpostes. (log currency/job)	fn_apply_analysis_sample_filters.do
Value added per hour	lnvaphrs	Natural log of vaht/tothrs. (log currency/hour)	fn_apply_analysis_sample_filters.do
Inputs per job	lninputspl	Natural log of inputs/totpostes. (log currency/job)	fn_apply_analysis_sample_filters.do
Inputs per hour	lninputsphrs	Natural log of inputs/tothrs. (log currency/hour)	fn_apply_analysis_sample_filters.do
Capital per job	lnkpl	Natural log of immocor/totpostes. (log currency/job)	fn_apply_analysis_sample_filters.do
Capital per hour	lnkphrs	Natural log of immocor/tothrs. (log currency/hour)	fn_apply_analysis_sample_filters.do
Investment per job	lninvcorppl	Natural log of invcorp/totpostes. (log currency/job)	fn_apply_analysis_sample_filters.do
Investment per hour	lninvcorphrs	Natural log of invcorp/tothrs. (log currency/hour)	fn_apply_analysis_sample_filters.do
TFP, fixed-share method, jobs	L_PROD_FS_postes	Log revenue productivity residual using the fixed-share production-function specification and job count as labor input. (log productivity)	fn_estimate_tfp.do
TFP, fixed-share method, hours	L_PROD_FS_hours	Log revenue productivity residual using the fixed-share production-function specification and hours as labor input. (log productivity)	fn_estimate_tfp.do
TFP, OP method, jobs	L_PROD_OP_W	Log revenue productivity residual from the retained OP-style estimation using job count as labor input. (log productivity)	fn_estimate_tfp.do
TFP, OP method, hours	L_PROD_OP_H	Log revenue productivity residual from the retained OP-style estimation using hours as labor input. (log productivity)	fn_estimate_tfp.do
Industry fixed-effect group	gmapetyr	Group of modal firm industry code (APET) by year. (categorical group)	fn_apply_analysis_sample_filters.do
Employment-zone fixed-effect group	gmze1990yr	Group of modal 1990 employment zone by year. (categorical group)	fn_apply_analysis_sample_filters.do
Cluster group	gmapetmze1990yr	Industry x employment-zone x year cluster identifier used in baseline descriptive/regression sampling. (categorical group)	fn_apply_analysis_sample_filters.do
1993 incumbent indicator	D_1993	Indicator that the 2006 firm is matched to the 1993 BTS-Entreprises/DADS firm file. (binary)	fn_compute_calibration_moments.do

Paper concept	Code variable	Definition / allowed values	Construction/source
New-firm share	sharenewfirms	Share of 2006 calibration-sample firms not matched to 1993. (share)	fn_compute_calibration_moments.do
Low-wage-bin employment share	share_postes_mw11 / share_hrs_mw11	Share of jobs/hours in the submitted low-wage bins around the minimum wage. (share)	fn_compute_calibration_moments.do
FARE firm identifier (Appendix G)	siren	Firm identifier used to deduplicate each FARE year and construct the 2010-2016 balanced panel. (identifier)	code/Programs/7_SAS_APPENDIX_G/01_compute_table_g1_zero_layer_moment.sas
FARE salaried employment FTE (Appendix G)	REDI_E200	Salaried employment in full-time equivalents; missing values are set to zero exactly as in the historical SAS calculation. (FTE)	code/Programs/7_SAS_APPENDIX_G/01_compute_table_g1_zero_layer_moment.sas
FARE historical selection variable (Appendix G)	REDI_R401	Variable used by the historical program for the annual inclusion rule: retain nonmissing REDI_R401 >= 15. Current CASD documentation labels REDI_R401 as total sales of merchandise; the modernization preserves the historical filter without reinterpretation. (source units; historical cutoff >= 15)	code/Programs/7_SAS_APPENDIX_G/01_compute_table_g1_zero_layer_moment.sas
0-layer firm / self-employment proxy (Appendix G)	self	Indicator equal to one for a firm present in every FARE year 2010-2016 whose sum of REDI_E200 over the seven years is zero. The reported target is mean(self). (0/1; reported as share)	code/Programs/7_SAS_APPENDIX_G/01_compute_table_g1_zero_layer_moment.sas